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Identifying ridehail riders through pairing ActivitySim and BEAM

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Abstract

The advent of on-demand transport modes such as ride-hailing and microtransit has challenged forecasters to develop new methods of forecasting the use and impacts of such modes. In particular, there is some professional disagreement about the relative role of activity-based transportation behavior models — which have detailed understanding of the person making a trip and its purpose — and multi-agent demand simulations which may have a better understanding of the availability and service characteristics of on-demand services. A particular question surrounds how the relative strengths of these two approaches might be successfully paired in practice. Using daily plans generated by the activity-based model ActivitySim as inputs to the BEAM multi-agent simulation, we construct nine different methodological combinations by allowing the choice to use a pooled ride-hail service in ActivitySim, in BEAM with different utility functions, or in both. Within each combination, we estimate ride-hailing ridership and level of service measures. The results suggest that mode choice model structure drastically affects ride-hailing ridership and level of service. In addition, we see that multi-agent simulation overstates the demand interest relative to an activity-based model, but there may be opportunities in future research to implement feedback loops to balance the ridership and level of service forecasts between the two models.

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1. Introduction

On-demand microtransit and ride-hailing have some potential to improve sustainability in car-dominated societies [29]. Some potential benefits include lower vehicular emissions, lessened roadway congestion, increased health, and economically viability [5, 22]. Unfortunately, forecasting the ridership and level of service of ride-hailing and other novel modes is a challenging task with no clear methodological consensus among practitioners. Individuals using ride-hailing vehicles behave differently than individuals using typical car modes of driving and carpooling, so under-

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standing behavior and the ride-hailing service capabilities is particularly challenging [8, 9, 16, 20]. These challenges echo those in modeling bike share [4, 15], e-scooters, or other mobility devices [14, 19, 18, 33].

Many different modeling methodologies exist with the purpose of better understanding the behavior of individuals using ride-hailing vehicles and other novel modes. Direct demand models are attractive for examining the spatial distribution and total use of novel modes [6, 14, 15], but are limited in their ability to examine responses to large-scale policies across modes or behaviors. A more common tool is an tour-based or activity-based model, which constructs daily activity patterns for simulated individuals and households and has been used to model the uptake of new modes [21, 24, 31]. Other forecasters use multi-agent simulation, which focuses on modeling the interactions between different agents, to understand the level of service of transport technologies [12, 27, 28]. Each of these model frameworks have methodological strengths and weaknesses that might lend themselves to being paired or combined. For example, an activity-based model has extensive information on the daily plans of the simulated agents including an understanding of a trip hierarchy, e.g. with mandatory trips shaping the options allowable on discretionary ones [10, 17, 25]. At the same time, activity-based models typically use a more rudimentary understanding of average travel times at a zonal level, an assumption that may be less valid with on-demand services where the level of service may depend on specific staging of limited resources and their use [26, 32]. Multi-agent simulations as largely developed have a symmetrically opposite challenge, with precise understanding of the locations and times of demand-responsive services [13, 3], but limited ability to anticipate the larger consequences of making mode or destination choices [7].

Some attempts have been made to combine activity-based models with multi-agent simulations. Zwick et al. [34] used MITO to estimate travel demand and MATSim to simulate that demand, estimating service criteria for pooled on-demand ride-hailing vehicles at a detailed level. Hörl et al. [11] discovered that when using a discrete mode choice model within MATSim, simulated agents made fewer irrelevant or impossible decisions, reducing runtime and improving realism. There is an open question, however, at the extent to which the two model frameworks can most profitably be intertwined. I.e., if the multi-agent simulation includes a mode choice component, must it be as detailed as the model used in the activity-based model?

In this research, we develop a series of experiments to understand the relative importance of a paired activity-based model and multi-agent simulation in forecasting the use of ride-hailing services. We do this by examining the ridership and level of service of ride-hailing predicted by different activity-based model and multi-agent simulation mode choice combinations. For each methodological combination, we estimate ride-hailing ridership and level of service outputs for a Salt Lake City, Utah case study region. The paper describes our methodological approach in the next section, followed by sections describing results of the experiments and discussing limitations alongside future research opportunities.

2. Methods

We developed a series of experiments to understand the relative importance of pairing an activity-based and multi-agent simulation in forecasting the uptake of ride-hailing. We used the Salt Lake City, Utah region as a case study for our experiments. We performed these experiments using ActivitySim [1] as the activity-based model and BEAM as the multi-agent simulation [2]. The ActivitySim implementation was imported from the Metropolitan Transportation Commission (San Francisco, California) implementation [23], but with travel time skims and a synthetic population representing the Salt Lake City region. The details of this implementation and calibration are described elsewhere [21]. A complete description of all the modeling parameters is not possible in this context, but all configuration files, model inputs, and plan outputs are available in a public online folder, with a public analysis code repository¹.

In this ActivitySim implementation, the utility V for person $n \in 1 : N$ choosing alternative mode $k \in K$ between origin zone $i \in I$ and destination zone $j \in J$ is defined as a linear combination of three sub-utilities for person

¹ The data input, model configuration, and output files can be found at <https://byu.box.com/v/tscore-mm03>, and the analysis code at https://github.com/cday97/mode_choice_thesis. Finally, our branch of the BEAM model is at <https://github.com/tscore-utc/beam>

attributes, path attributes, and location attributes as:

$$V_{nk} = V_{nk}^{Person} + V_{nk}^{Path} + V_{nk}^{Location} \quad (1)$$

The sub-utilities themselves are defined as follows, with the person utility

$$V_{nk}^{Person} = \alpha_{kn} + \beta_{kp}^1(cost_{k,ij}) + \beta_{kp}^2(age_n) + \beta_{kp}^3(hhsizen) \quad (2)$$

including an alternative specific constant α that varies by the household auto availability, the age of the individual, and household size; that is, the effect of these person-level variables on mode choice is captured by a differentiated alternative specific constant. In this way, the constant term itself serves as a personal attribute variable. The *cost* variable describes any monetary costs associated with the mode choice, including transit and ridehail fares, tolls, and an auto operating costs. This term is associated with the person attribute as ActivitySim's implementation uses a person-level value of time associated with the person's household income. The path utility

$$V_{nk}^{Path} = \beta_{kp}^4(dist_{k,ij}) + \beta_{kp}^5(tv_{k,ij}) + \beta_{kp}^6(te_k) + \beta_{kp}^7(tw_k) + \beta_{kp}^8(prox_k) + \beta_{kp}^9(xfer_k) \quad (3)$$

where, *dist* is distance, *tv* is vehicle travel time, *te* is egress time, *tw* is wait time, *prox* is proximity to transit, *xfer* is number of transfers. The final utility equation describes the attribute of the locations at the end of the trip,

$$V_{nk}^{Location} = \beta_{kp}^{10}(ZDI_{k,i}) + \beta_{kp}^{11}(ZDI_{k,j}) + \beta_{kp}^{12}(ZTI_{k,j}) + \beta_{kp}^{13}(CBD_{k,j}) \quad (4)$$

with *ZDI* a zonal density index, *ZTI* is zonal topography index related to the grade of the hills in the zone, *CBD* is a central business district. In all equations the terms $\beta_p^1 : \beta_p^{13}$ are estimated coefficients that vary by tour purpose *P*.

As shown in Equation (1), ActivitySim uses the combination of person, path, and location variables to calculate the mode choice alternative. The combination of these different variable types determines whether or not a person selects a ride-hailing mode. But the travel times in Equation 3 are exogenously provided in the travel time skims rather than determined based on the equilibrium between supply and demand.

The standard BEAM implementation uses a simple multinomial logit mode model to determine which mode any particular agent will use on any particular trip. This model calculates the utility *V* for person $n \in 1 : N$ choosing alternative mode $k \in K$ between origin zone $i \in I$ and destination zone $j \in J$ as:

$$V_{nk} = \alpha_k + \beta_k^1(cost_{k,ij}) + \beta_k^2(tv_{k,ij}) + \beta_k^3(xfer_k) \quad (5)$$

where, α is the alternative specific constant that varies by mode, *cost* is the fare associated with the travel mode in BEAM, *tv* is vehicle travel time, *xfer* is number of transfers and $\beta_p^1 : \beta_p^3$ are estimated coefficients that vary by mode. However, we improved the BEAM's default mode choice model to recognize the tour purpose as specified in the input plans. This in turn allowed us to implement the same utility equations in BEAM that ActivitySim uses, which differ by trip purpose *P* and the attributes described in Equations 2 through 4. We did not change BEAM's logic for identifying available modes, or for allocating vehicles to subsequent trips. This means that if an agent selects a ridehail option within BEAM, they are more likely to use ridehail on subsequent trips as their vehicle is no longer with them.

This research design allows us to generate nine scenarios with different combinations of ridehail sensitivity in the two models. With regards to ActivitySim, two different methods are possible: we can omit ridehail as an alternative in

Table 1. ActivitySim-to-BEAM Mode Choice Combination Scenario Names

		ActivitySim Implementation	
		Plans without ride-hail	Plans with RideHail
BEAM Implementation	None		AsimRideHail
	RideHail	BeamRideHail:Path BeamRideHail:PPL	AsimBeamRideHail:Path AsimBeamRideHail:PPL
	All	BeamAll:Path BeamAll:PPL	AsimBeamAll:Path AsimBeamAll:PPL

the mode choice model, or we can include it. The former option then relies entirely on the endogenously determined availability of ridehail vehicles in the BEAM multi-agent simulation. In BEAM, there are three options for the ridehail architecture:

1. *None*: the decision to use all modes is determined by the daily plans supplied from ActivitySim.
2. *Ridehail*: BEAM determines whether a trip will happen on ridehail instead of the supplied mode.
3. *All*: Beam re-determines all mode choices from the supplied plans.

The combination of these two dimensions would lead to six scenarios, though the option where neither ActivitySim nor BEAM allows ridehail as an option is not a useful analysis in this context. A third dimension is that within BEAM, two different mode choice utility expressions can be used: a form with only the path terms (Equation 3), or a form with all terms for person, path, and location (PPL, Equation 1). Allowing these two options on the four scenarios with BEAM mode innovation leads to an additional set of four scenarios. The resulting nine scenarios are charted in Table 1.

3. Results

We ran each of the nine scenarios in BEAM with a random 15% of households drawn from the ActivitySim plans. Each scenario was run for only 12 iterations to facilitate a detailed analysis of behavior evolution among the simulated agents.

Table 2 shows the number of estimated trips for the ride-hail, pooled ride-hail, and ride-hail transit type modes for all nine mode choice combinations. Table 2 also includes the number of forecasted trips within the plans with ride-hail created by ActivitySim, before the mode choices were changed by BEAM. Lastly, Table 2 provides a total percentage column, showing the total mode share ridehail trips relative to all trips in the scenario.

By comparing scenarios against each other, we can understand how slight differences in the mode choice model can significantly affect estimated ridership totals. We first compare the ActivitySim scenario with the AsimRideHail scenario. The ActivitySim scenario represents the ride-hailing input plans created by ActivitySim before being inserted into BEAM and the AsimRideHail scenario represents how these ride-hailing input plans change in BEAM while undergoing no new mode choice. Comparing these two scenarios we see the estimated 4,249 ride-hailing trips of ActivitySim diminished to 300 trips within BEAM because the AsimRideHail scenario was unable to recreate the same ride-hailing paths as ActivitySim. Then, comparing the *RideHail* and *All* type scenarios with the AsimRideHail scenario and ActivitySim scenario we see that when BEAM mode choice innovation is turned on for all or part of the agents, BEAM predicts significantly higher ridership totals. As a result, we suppose that BEAM is prone to estimating

Table 2. Estimated Ridehail Ridership by Mode Choice Scenario

Scenario Name	Ride-hail	Pooled Ride-hail	Ride-hail to Transit	Total	Total (%)
<i>ActivitySim - Inputs to BEAM</i>	2412	1837	0	4249	0.38
AsimRideHail	269	31	0	300	0.03
BeamRideHail:Path	45001	25014	2943	72958	6.18
BeamRideHail:PPL	18907	38935	4621	62463	5.34
AsimBeamRideHail:Path	21519	40873	5437	67829	5.75
AsimBeamRideHail:PPL	47422	27327	3848	78597	6.60
BeamAll:Path	4671	1596	38	6305	0.57
BeamAll:PPL	3146	6366	121	9633	0.86
AsimBeamAll:Path	3470	4596	90	8156	0.73
AsimBeamAll:PPL	3153	6031	156	9340	0.83

higher ridership totals for ride-hailing modes than ActivitySim when modal innovation is turned on, and lower totals when turned off.

We also notice that different BEAM mode choice models predict different forecasted ride-hailing ridership levels. The *None* type scenario (AsimRideHail) assigns few agents to the ride-hail mode. The *RideHail* type scenarios produce the largest number of ride-hail modes among BEAM mode choice structures. The *All* type scenarios produce more ride-hail modes than the *None* type, but less ride-hail modes than the *RideHail* type. In addition, the *All* type scenarios predicts similar ridership values as ActivitySim. The number of ride-hail only type trips is 2,259 more in the BeamAll:Path scenario than in the ActivitySim scenario. All the *All* type scenarios as well as ActivitySim predict under 1% total ride-hailing modes. Overall, forecasted ridership is affected significantly by which mode choice structure is used by BEAM; this conclusion is clear.

Figure 1 shows a detailed distribution of wait times for ride-hailing vehicles for the scenarios. As with ridership, we compare the scenarios with the *Asim* prefix against the scenarios without, and see only a slight difference in maximum wait times. Scenarios AsimBeamRideHail:PPL, AsimBeamAll:Path, and AsimBeamAll:PPL have almost identical mean wait times when compared to their counterparts in BeamRideHail:PPL, BeamAll:Path, BeamAll:PPL. We suppose that the existence of ride-hail in the initial plans will not affect *most* ride-hail wait times.

Alternately, comparing different BEAM mode options does significantly affect ride-hail wait times. The *None* type scenario (AsimRideHail) has the largest spread of wait times, the lowest mean wait time, and is “bottom heavy” – referring to the fact that a major cluster of users wait less than 7.5 minutes. The *All* type scenarios have higher mean wait times (approx. 9.3 minutes) than the *None* type and lower mean wait times than the *RideHail* type. Neither top nor bottom heavy, the *All* type scenarios have a more even spread in wait times, ranging from 0.4 to 21 minutes. BEAM seems to paint ride-hail alternatives as more desirable than ActivitySim, as more users are willing to wait longer (12 to 18 minutes in the *All* scenarios). This is especially true with the *RideHail* type scenarios, as a large cluster of users are willing to wait 7.5 to 20 minutes. The *RideHail* type scenarios have the largest mean wait times (approx. 10.65 minutes). By simply viewing the wait time distribution, we see a difference in level of service depending on model structure. Wait time is significantly affected by which mode choice structure is used by BEAM, just like as was concluded with ridership.

We also conducted a statistical comparison between quantile ranges between all three model structures. Using a quantile test based on the method proposed in Wilcox et al. [30], a significant statistical difference exists between the *None* and *All* models, the *All* and *RideHail* models, and the *None* and *RideHail* models at the 20th and 50th percentile levels. At the 80th percentile level, a significant difference exists between the *All* and *RideHail* models and the *None* and *RideHail* models, but not between the *None* and *All* models. The results of these statistical tests suggest that a different level of service will result based on which mode choice model structure is used. We also ran quantile tests between like model structures of scenarios with and without ride-hailing in the input plans (i.e., BeamRideHail:Path and AsimBeamRideHail:Path) and noticed no statistical difference at the 20th, 50th, and 80th percentiles. These

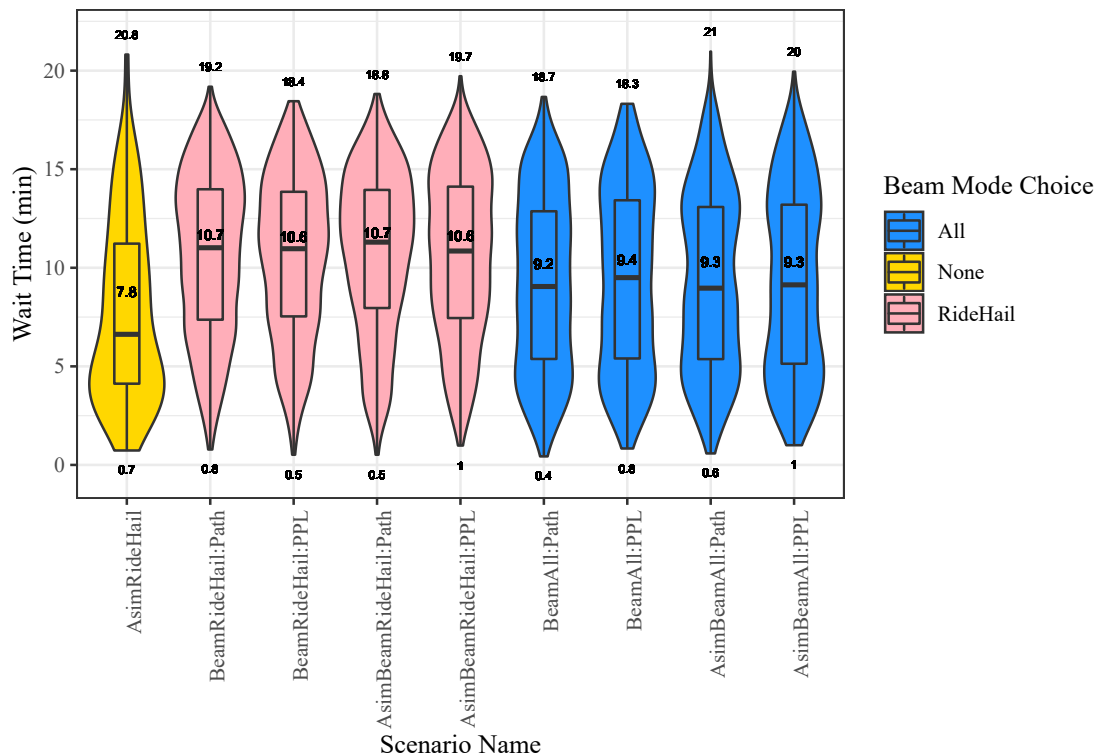


Fig. 1. Estimated wait time distribution in scenarios.

statistical tests indicate that wait times will not change significantly based on the inclusion of ride-hailing in the input plans.

The last group to compare collectively is between the Path and PPL models. By comparing BeamRideHail:Path and BeamRideHail:PPL, BeamAll:Path and BeamAll:PPL, and AsimBeamAll:Path and AsimBeamAll:PPL, we see that the Path models estimate a slightly higher maximum wait time. In addition, BeamRideHail:Path and AsimBeamRideHail:Path seem to have a larger cluster above a 10 minute wait time than BeamRideHail:PPL and AsimBeamRideHail:PPL. Besides these two observations though, the differences between utility parameters is minimal. Although ridership was affected by which utility parameters were used, wait time is only slightly affected.

4. Discussion and Conclusion

There are several limitations in this research design. First, the BEAM model was under active development during the process of this research, which resulted in some difficulties with calibration and identification of available autos for trips. Second, using a research implementation of a travel demand model rather than the region's accepted model means that overall mode splits were not necessarily representative of extant travel patterns.

The purpose of this research is not, however, a test of model accuracy in estimating ride-hailing ridership and level of service but rather a high-level comparison between different modeling approaches. With this comparison we aim to understand the relative effect different paired mode choice combinations can have in forecasting ride-hailing services.

What this research has shown is that there are meaningful differences resulting from where ridehail mode choice is made. The fact that many of the agents identified as ridehail passengers by ActivitySim could not find or use ridehail in BEAM is not surprising given the relative information available to agents in the two models. Allowing for BEAM to determine some of the mode choices is likely appropriate. But it is difficult to believe that the inflated shares obtained when all BEAM agents can choose ridehail or their existing mode are going to be realistic. This is particularly true given that at the moment of choice, BEAM (or MATSim) does not have a good picture of the future trips that the

person will make. It is possible that the convergent behavior of existing multi-agent systems may cause agents to discard plans where choosing ridehail was not optimal. But given the strong time and potentially even cost benefits of ridehail vis a vis transit, it may be better to heuristically limit the options available to BEAM agents based on more information than just the purpose of the trip and its origin and destination characteristics. Activity-based models are more properly set up to handle this. Another possible solution would be to use the multi-agent simulation to build a better picture of the ridehail service capabilities, better informing the plans made in the activity-based model. Further research and experimentation is required.

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